

A TASK IS USABLE ONLY WHEN IT GETS DONE

A visitor without a smartphone can speak to staff and leave with paper. One opening shift now has visible workload and stop deadlines.

01 ASK · 02 HUMAN CHECK · 03 LEAVE WITH PAPER

02 ASSISTED OPEN



AI organizes public material; staff verify and issue paper

03 MANUAL-ONLY DEGRADED



network and interfaces off, staffed paper service continues

6 / 0

QUOTE CARDS / QUOTES

total and shift count null

8 / 21.5

PUBLIC / FUNCTIONAL HOURS

not people or roster

5 / 6

STOP TYPES / FAULT CARDS

six event deadlines

90 / 0

SYNTHETIC / FIELD

concept is not field proof

SIXTEEN PAGES, SEVEN DECISIONS

Every dimension pairs a claim with locatable evidence and a boundary.
A juror can inspect the One-Shift Opening Book before moving to space, public tasks, and field gaps.

01	BRIEF FIT	04 / 06 / 13	three scales + two service chains
02	ORIGINALITY	01 / 03 / 08	Works Bay + One-Shift Opening Book
03	AI INNOVATION	03 / 12	human first + event deadlines
04	IMPLEMENTATION	03 / 08 / 15	effort, quote cards, stop + restore
05	PUBLIC INTEREST	10 / 11 / 13	same task without an app
06	RISK + ETHICS	03 / 12 / 16	quotes and field results remain separate
07	EXPRESSION	ALL / 15 PAIRS	pairwise bilingual audit; boundary on-page

Fifteen Chinese-English pairs received an author semantic audit. Independent translation review and public-comprehension testing remain incomplete.

WHAT WORKLOAD DOES ONE OPENING SHIFT REQUIRE

Eight public hours, 21.5 functional-role hours, seven checkpoints, and six quote cards. Supplier quotations and field results remain zero.

JING-ZHANG WORKS · ONE-SHIFT OPENING BOOK · V1.9

WRITE THE CLOCK, WORKLOAD, AND STOP RULE BEFORE OPENING

EIGHT REFERENCE PUBLIC HOURS · 21.5 FUNCTIONAL-ROLE HOURS · SEVEN CHECKPOINTS · SIX QUOTE CARDS · ZERO SUPPLIER QUOTATIONS

SEVEN CHECKPOINTS THAT MUST LEAVE EVIDENCE

T-60	T-30	T-10	T+00	T+120	T+240	T+480
walk full route	replay on paper	sign release	show state	independent replay	run fault card	close + reconcile

FIVE FUNCTIONAL WORKLOADS

duty lead	9.75h	route through close
paper service	9h	paper task + takeover
access replay	1h	pre-open + midday
safety fault	1h	stop + fault card
information + data	0.75h	source + deletion
21.5 h functional coverage, not people, roster, wages, or legal working time		

SIX EVENT DEADLINES

01	before next public transaction	safety / privacy / critical information / network
02	before paper is issued	staff accepts human takeover
03	before visitor leaves	give issue ID + review date
04	before closing sign-off	reconcile issues, versions, deletion
05	before public reopening	recovery evidence + independent replay
06	after supplier exit	disconnect, export allowed records, restore

SIX QUOTE-READY WORK CARDS

C01 6 routes + 12 observations	C02 3 paper sets + 1 master	C03 one 6.0 × 4.8m set	C04 one test at a time	C05 21.5 functional hours	C06 permanent works zero
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REFERENCE WORK ORDER · 0 SUPPLIER QUOTATIONS · TOTAL NULL · SITE, DATE, PEOPLE, SHIFT COUNT, PERMITS, AND FIELD RESULTS EMPTY

THREE SCALES, EACH ENDING IN A HUMAN STEP

Research connects industry and regional resources; overall design organizes walking, public space, and service; key areas turn decisions into reversible prototypes.

RESEARCH
Industry and regional loops

KEY AREAS
Reversible spatial operations

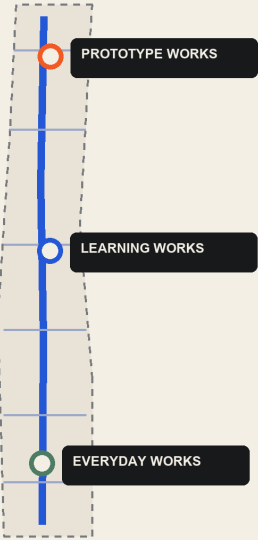
OVERALL
Land, mobility, public service

RETURN
Problems return to R&D and maintenance

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HOW ONE TASK GETS DONE

A problem moves through Prototype, Learning, and Everyday Works; failure returns through the second service chain



TWO SERVICE CHAINS

LAB TO STREET



PROBLEM TO CHANGE



ANY FAILED STEP REACHES A PERSON AND PAPER

LEARN METHODS FROM SIX CASES, NOT THEIR LOOK

Shared platforms, continuous feedback, agile pilots, resident validation, liveability metrics, and evaluation before scale are rewritten as Jing-Zhang adoption rules. No case imagery or brand asset is copied.



Qinghe Railway Station West Square; arrival, crossing, and station-city interface reference
PHOTO-09 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Accessible ramp at Tuqiao Station; a continuous-ramp reference from another Beijing location
PHOTO-14 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Guidance sign in Beijing Railway Station; multilingual and continuous-wayfinding reference
PHOTO-15 · Siyuwj · CC BY-SA 3.0 · External context reference; not field evidence

SHARED
Punggol

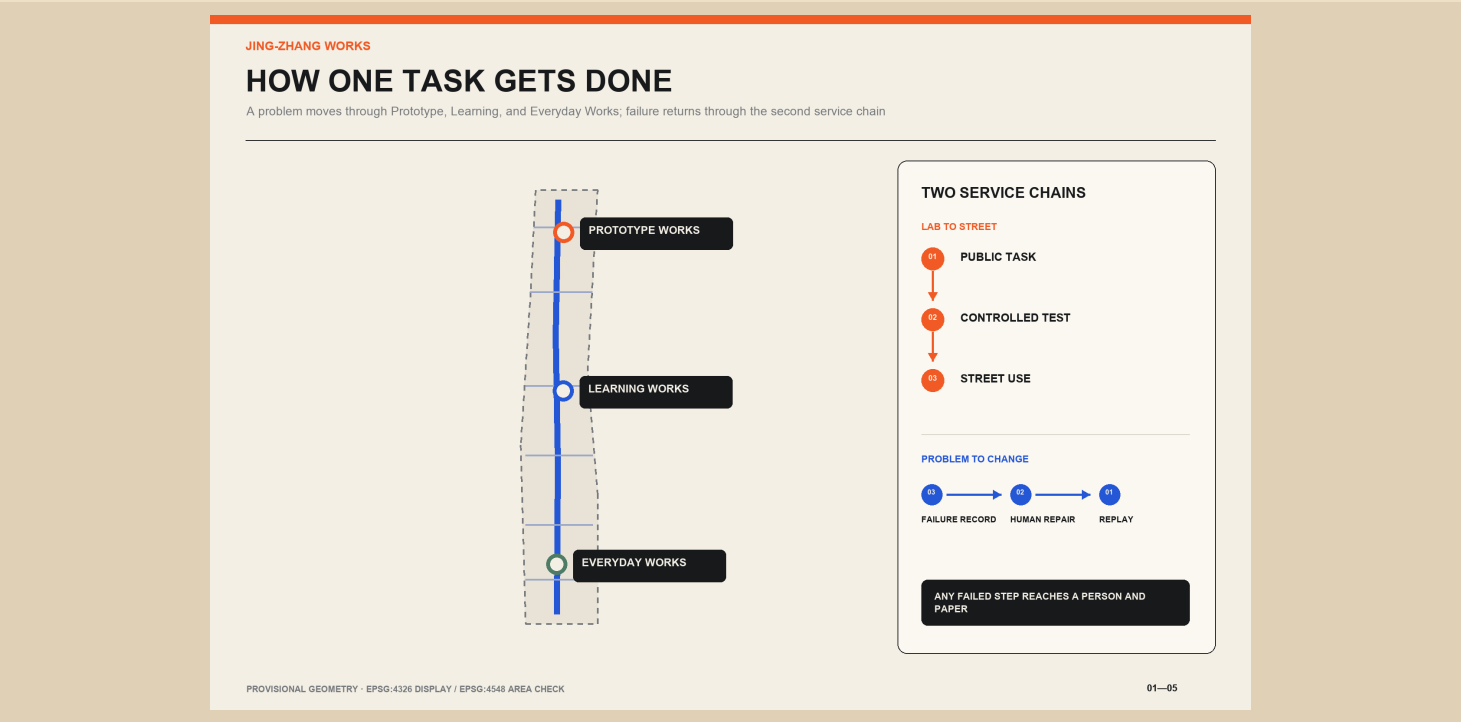
AGILE
Kalasatama and Paris

REAL USE
Woven City and Magok

PUBLIC RIGHT
Barcelona liveability

THREE WORKS, TWO CHAINS THAT CAN TURN BACK

Lab to Street controls release. Problem to Change controls maintenance, suspension, and retirement. Technology must be able to return when it fails.



PROTOTYPE
Controlled test before street

LEARNING
Lower service barriers

EVERYDAY
Face commute and daily life

WINGS
Tech service and scenarios



Locomotive display in Jingzhang Railway Park; industrial-culture and test-space context
PHOTO-06 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Student reading room at the University of Science and Technology Beijing; shared-learning and night-study reference
PHOTO-16 · Ccyber5 · Public domain dedication by the copyright holder · External context reference; not field evidence



North concourse of Line 13 Dazhongsi Station; everyday transit and staffed-service interface reference
PHOTO-11 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

PROTOTYPE WORKS: CONTAIN FAILURE FIRST

An observation foyer, accessible test lane, isolated test pocket, physical stop, and failure review make a controlled section. A robot that fails here never reaches the public street.

COMPONENT

C04 accessible test lane

OWNER

Test lead and site safety

RESOURCE

M-class movable kit

HARD STOP

Break approach or failed E stop

JING ZHANG WORKS

THREE WORKS, THREE SPATIAL PROTOTYPES

One service rule becomes controlled testing, public learning, and everyday service space

01 PROTOTYPE WORKS

OBSERVE ACCESS LANE TEST POCKET STOP + REVIEW

MOVABLE PROTOTYPE: SETTING TO VERIFY

02 LEARNING WORKS

ARRIVE ASK FIRST LEARNING TABLE HUMAN CHECK

MOVABLE PROTOTYPE: SETTING TO VERIFY

03 EVERYDAY WORKS

SHADED ENTRY QUIET REST TAKEOVER CHANGE LOG

MOVABLE PROTOTYPE: SETTING TO VERIFY

PROVISIONAL GEOMETRY SPSS-K333 DISPLAY SPSS-K346 AREA CHECK

01-05

Locomotive display in Jingzhang Railway Park; industrial-culture and test-space context
PHOTO-06 · N599FZ · CC-BY-SA 4.0 · External context reference; not field evidence

PROVISIONAL GEOMETRY · FIELD PERFORMANCE UNKNOWN

official_boundary=false · concept only

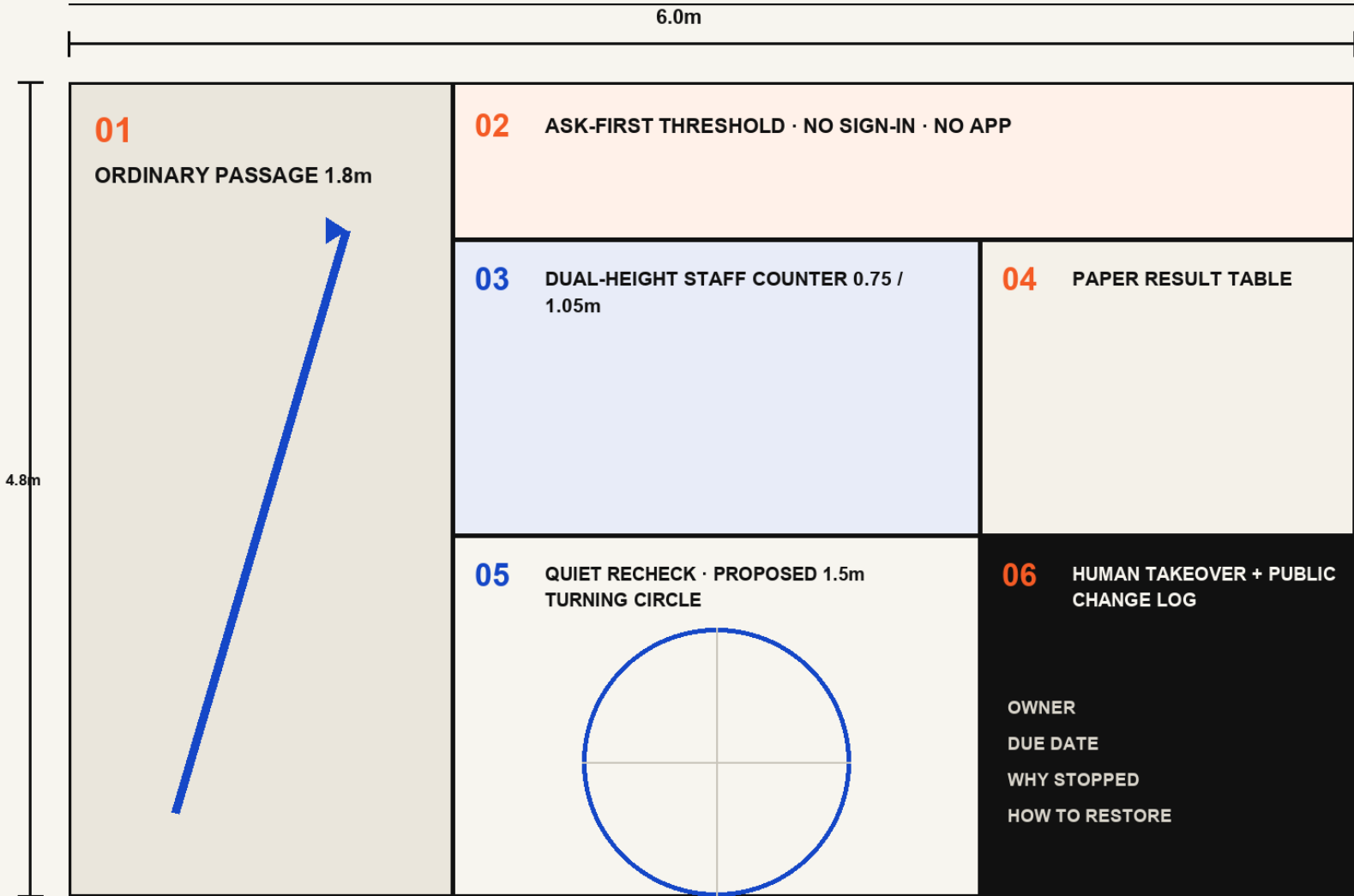
BUILD THE WAY OUT INTO THE ROOM

6.0 × 4.8 metres is a trial-assembly dimension, not a site survey or compliance finding. No public installation before site, fire, egress, accessibility, and operating approvals.

JING-ZHANG WORKS · 1:1 TRIAL ASSEMBLY

WORKS BAY

A public-service trial room that can answer, hand over, print a result, and close immediately when the service fails



FOUR OPERATING STATES

01	MANUAL OPEN works without AI
02	ASSISTED OPEN AI advises only
03	MANUAL-ONLY DEGRADED equivalent route remains
04	CLOSED FOR RECOVERY stop, delete, restore

28.8 m² TRIAL DIMENSION · NOT A SITE MEASUREMENT OR COMPLIANCE FINDING

6 SPATIAL MOVES 4 OPERATING STATES 5 HUMAN SIGN-OFFS 0 FIELD RESULTS

DESIGN PACKAGE · NOT SITED · NOT AUTHORISED · NOT RUN

EVERYDAY WORKS: SURVIVE AN ORDINARY EVENING

A shaded entry, quiet rest, human takeover, and public change record form an everyday section. Dazhongsi siting remains unverified, so every component stays movable.

COMPONENT
C02 Takeover + C03 Change Log

OWNER
Safety lead and service owner

RESOURCE
S/M movable kit

HARD STOP
Crowd, error, or blocked exit



Accessible lift at Dazhongsi Station Exit E; step-free arrival reference
PHOTO-12 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

North concourse of Line 13 Dazhongsi Station; everyday transit and staffed-service interface reference
PHOTO-05 · N4077 · CC BY-SA 4.0 · External context reference; not field evidence

NOT PERSONAS: EIGHT MOMENTS WHERE A TASK BREAKS

No phone, wheelchair, low vision, care work, neighbourhood life, study, testing, and visiting from abroad each map to a complete, walkable task—not an abstract avatar.



Accessible elevator at Dajiaoting Station; a service-facility reference from another Beijing location
PHOTO-13 · Wcr1993 · CC BY-SA 4.0 · External context reference; not field evidence



Accessible ramp at Tuqiao Station; a continuous-ramp reference from another Beijing location
PHOTO-14 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Guidance sign in Beijing Railway Station; multilingual and continuous-wayfinding reference
PHOTO-15 · Siyuwj · CC BY-SA 3.0 · External context reference; not field evidence



Student reading room at the University of Science and Technology Beijing; shared-learning and night-study reference
PHOTO-16 · Ccyber5 · Public domain dedication by the copyright holder · External context reference; not field evidence

EIGHT PUBLIC TASKS, EACH WITH A NO-APP ROUTE

Routes, language, shade, night study, older-adult service, sensory access, jobs, and crowd passage share one floor: a person can find staff and leave with paper.



Qinghe Railway Station West Square; arrival, crossing, and station-city interface reference
PHOTO-09 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



South underground corridor of Qinghe Railway Station; continuous wayfinding and walking-interface reference
PHOTO-10 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Accessible lift at Dazhongsi Station Exit E; step-free arrival reference
PHOTO-12 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

MOBILITY
Arrival, shade, crowd

SERVICE
Older, language, sensory access

GROWTH
Night study, jobs, events

COMMON RULE
Same task without an app

FOUR INDUSTRY TESTS THAT FIRST LEARN TO STOP

Robots, closed delivery, privacy sensing, and digital twins stay controlled. Boundary breach, human proximity, unclear data purpose, or failed command triggers a hard stop.

ROBOT
Parallel access and recovery
DELIVERY
Closed-zone speed and takeover
SENSOR
Aggregate and delete

TWIN
Synthetic data and human command

Locomotive display in Jingzhang Railway Park; industrial-culture and test-space context
PHOTO-06 · N509FZ · CC-BY-SA 4.0 · External context reference; not field evidence

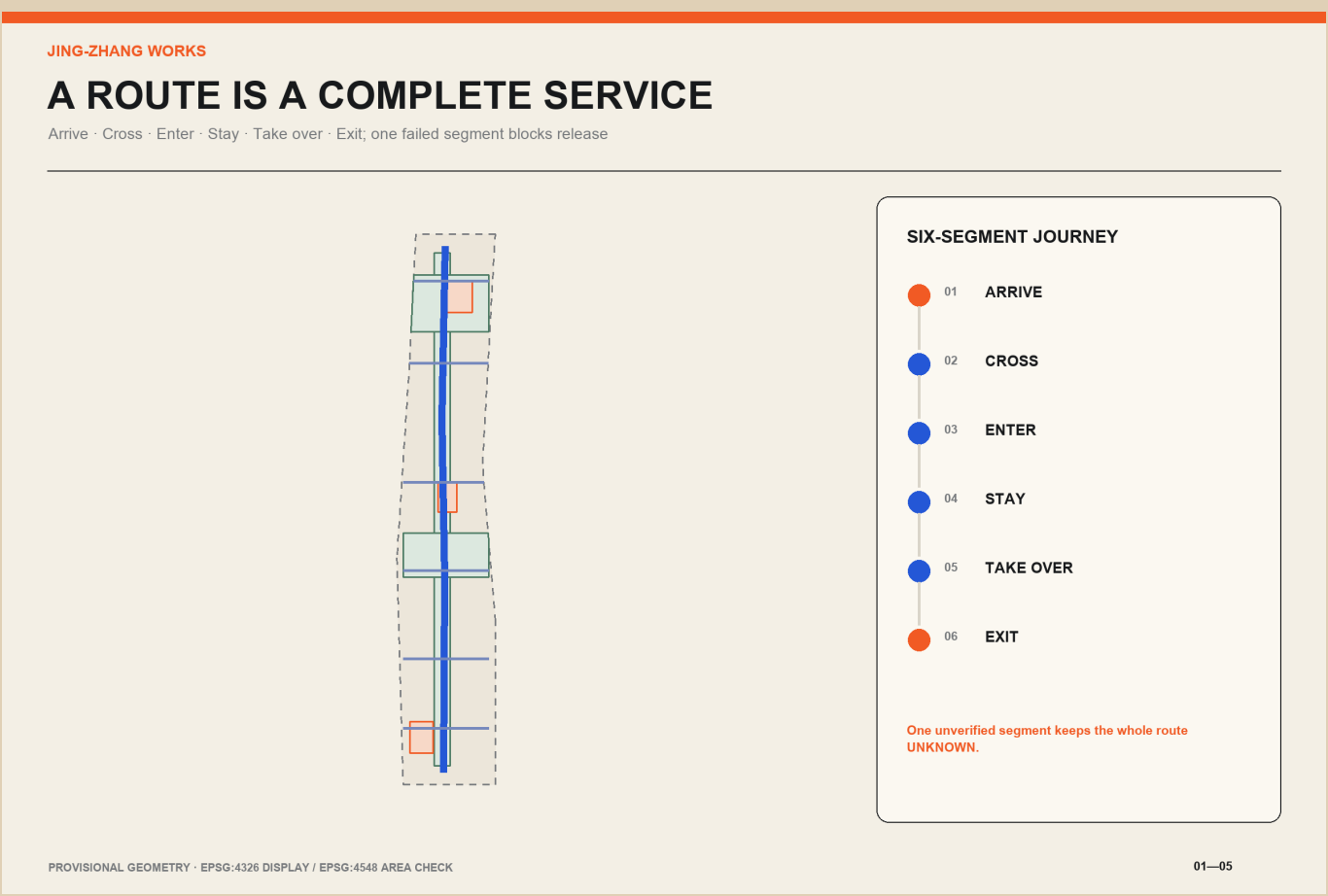
PROVISIONAL GEOMETRY · FIELD PERFORMANCE UNKNOWN

South portal of the Badaling tunnel, 1909 or earlier; engineering-history context
PHOTO-02 · 1876-1915 · Public domain in China · External context reference; not field evidence

official_boundary=false · concept only

ONE ROUTE COUNTS FROM ARRIVAL TO EXIT

Arrival, crossing, entry, stay, takeover, and exit form one service. One unverified segment blocks the route; a polished fragment never cancels a broken middle.



South underground corridor of Qinghe Railway Station; continuous wayfinding and walking-interface reference
PHOTO-10 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

01
ARRIVE

02
CROSS

03
ENTER

04
STAY

05
TAKEOVER

06
EXIT

LAND EXPLAINS RELATIONSHIPS; BUILDINGS KEEP THEIR UNKNOWN

Six concept zones check topology and task relationships inside the provisional frame. FAR, height, ownership, structural safety, and building action wait for professional evidence.

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PROGRAMME FOLLOWS PUBLIC TASKS

Six concept zones cover the provisional frame; statutory use and intensity stay unknown

Research and development testing

Qinghe and continuous green space

Learning and innovation services

Industry and public services

Community and culture services

Culture and civic welcome

EIGHT-RESOURCE ORDER

01 LAND

02 SPACE

03 INDUSTRY

04 FUNDING

05 TALENT

06 COMPUTE

07 DATA

08 SCENARIOS

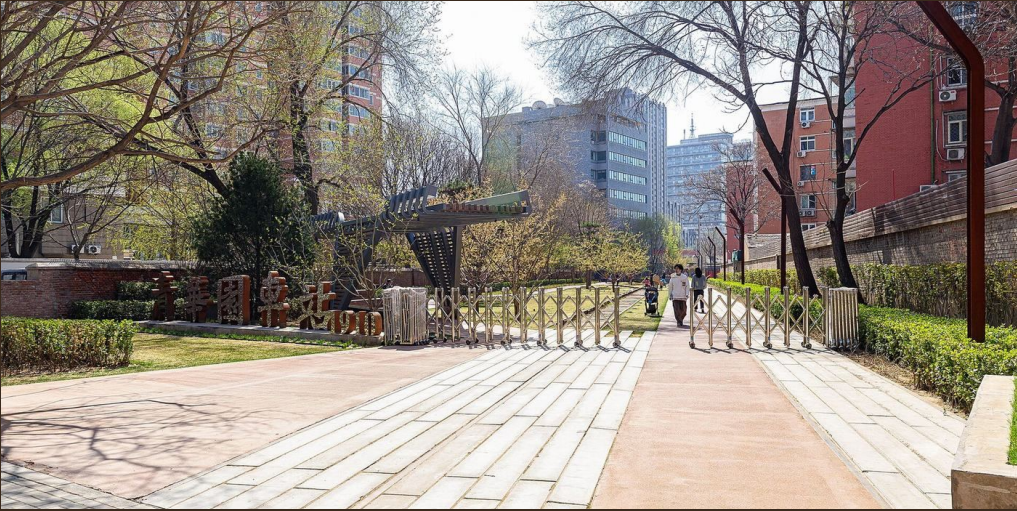
A resource is unsecured until authorization, quantity, term, and exit are recorded.

PROVISIONAL GEOMETRY · EPSG:4326 DISPLAY / EPSG:4548 AREA CHECK

01—05



Old Qinghuayuan Station after renovation; adaptive-reuse reference
PHOTO-04 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



North entrance of Old Qinghuayuan Station Park; railway heritage entering daily public space
PHOTO-05 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

LAND	RESOURCES	KNOWN	UNKNOWN
Six-zone topology coverage 1.0	Eight-item adoption order	Concept layers and footprints	Statutory use, FAR, height, building action

ONE PAPER HANDS THE NEXT STEP TO A PERSON

A bilingual blank template, not a real service result. AI drafts only from dated public sources; staff verify and sign every critical fact.

JING-ZHANG WORKS · U05-A · BLANK HANDOVER

PAPER RESULT RECEIPT

HAND PUBLIC INFORMATION TO THE NEXT OFFLINE ACTION; KEEP HUMAN DECISIONS WITH PEOPLE

01	TASK TO COMPLETE
02	PUBLIC SOURCE + PUBLICATION DATE
03	MATERIALS TO PREPARE
04	NEXT OFFLINE STEP
05	WHEN THIS PAPER MUST BE RECHECKED
06	HUMAN VERIFIER + TIME
07	ISSUE ID

AI MAY

- 1 extract candidate public fields
- 2 draft plain language + bilingual text
- 3 flag missing source or expiry

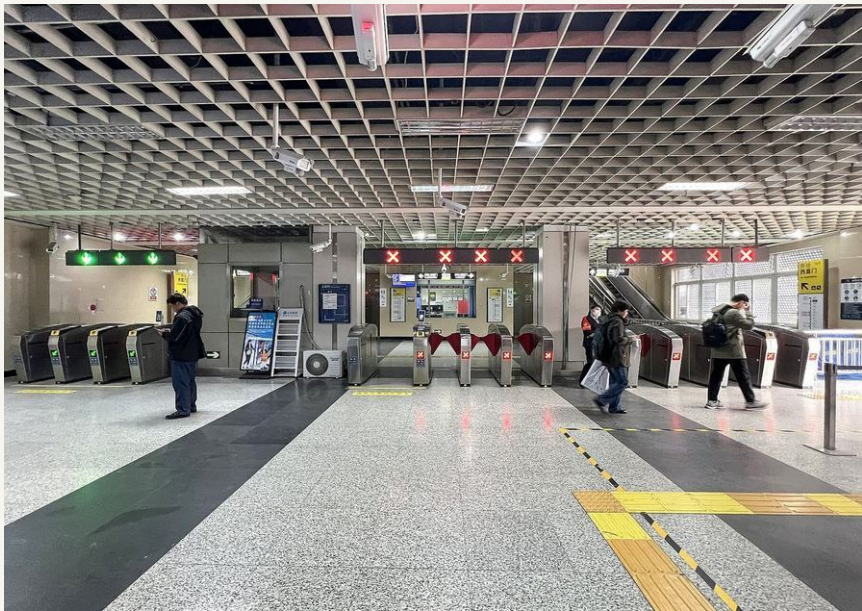
AI MUST NOT

- ✗ decide eligibility
- ✗ invent a source
- ✗ replace human sign-off
- ✗ retain a personal profile

NO ISSUE IF SOURCE, EXPIRY, OFFLINE STEP, OR HUMAN SIGN-OFF IS MISSING

BLANK TEMPLATE · NO REAL ANSWER · NO PERSON · NO SIGNATURE · NO TRANSACTION

15 / 16



North concourse of Line 13 Dazhongsi Station; everyday transit and staffed-service interface reference
PHOTO-11 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Guidance sign in Beijing Railway Station; multilingual and continuous-wayfinding reference

MISSING ONE OF FOUR MEANS NO ISSUE

12/12 RULES COMPLETE, 0/12 FIELD TASKS COMPLETE

Structural completeness does not mean the city works. Only a replay by a real user, operator, and qualified reviewer can turn unknown into a measured value.



Qinglongqiao Railway Station in 1909; Jing-Zhang historical context
PHOTO-01 · Public domain in China · External context reference; not field evidence

12/12
STRUCTURAL CONTRACTS READY

0/12
AUTHENTIC FIELD TASKS COMPLETE

Zhongguancun Plaza and street at dusk; public-life context in the innovation district
PHOTO-08 · Charlie fong · Public domain dedication by the copyright holder · External context reference; not field evidence